MATH 130: 10/20 WORKSHEET

A *data set* is a collection of data. Often these will be numbers, but they can be qualitative. When talking abstractly, we will list the data in a data set like

$$x_1, x_2, \ldots, x_n.$$

That is, there are n data in the data set and each datum x_i gets an index i for its position.

Given a data set it would be convenient to have a way to specify a typical or average member of the data set. There's a few useful ways to do this.

Mean.

Given a numerical data set $x_1, x_2, \ldots x_n$, their *mean*, commonly denoted as $\bar{x}$, measures their center of gravity. Imagine the number line as a balancing beam with a weight placed at the location of each x_i. The mean $\bar{x}$ is where you would place a fulcrum so that the two sides would be balanced. Mathematically, we describe it with the following formula:

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum x_i}{n}.$$

That is, the mean is the sum of the data set divided by the number of quantities in it.

The notation $\sum x_i$ is called *summation notation*, and means to sum up all the x_is. It's useful to have a shorthand for writing $x_1 + x_2 + \cdots + x_n$. The symbol $\sum$ is the capital Greek letter Sigma. Think $\sum$ is the Greek S, with S for "sum".

Here's another way to think about the mean. Imagine the data set plotted on the number line. If you want a position which minimizes the total distance to the data across the whole data set, that is the mean.

This is the most important notion of an average. It enjoys nice mathematical properties which enable more complicated uses of statistics.

Median.

Take a numerical data set $x_1, x_2, \ldots, x_n$. Their *median* is the value in the middle of you list them in order. That is, the median is the position where half the values are smaller and half are larger. To make this precise, let's assume the values $x_1, x_2, \ldots, x_n$ are listed in increasing order, and split into two cases.

- If n is odd, the median is the value exactly in the middle, namely $x_{(n+1)/2}$.
- If n is even, there isn't a single value exactly in the middle. So instead take the average of the closest two:
$$\frac{x_{n/2} + x_{n/2+1}}{2}.$$

Mode.

Take a data set $x_1, x_2, \ldots, x_n$, possibly numerical, possibly qualitative. Their *mode* is the value which occurs the most often in the list. If there is a tie, the data have multiple modes.

Here's one of the nice mathematical properties of the mean.

Linearity of the mean.

The mean enjoys the following two properties, collectively called *linearity*.

- If you multiply every datum in a data set by the same constant k, the new mean is k times the old mean.
- Suppose you mix two data sets $\{x_i\}$ and $\{y_j\}$ as follows. To make a new data set, you take every possible way to add values from the two data sets together, that is all possible $x_i + y_j$. The mean of this new data set is the sum of the two means.